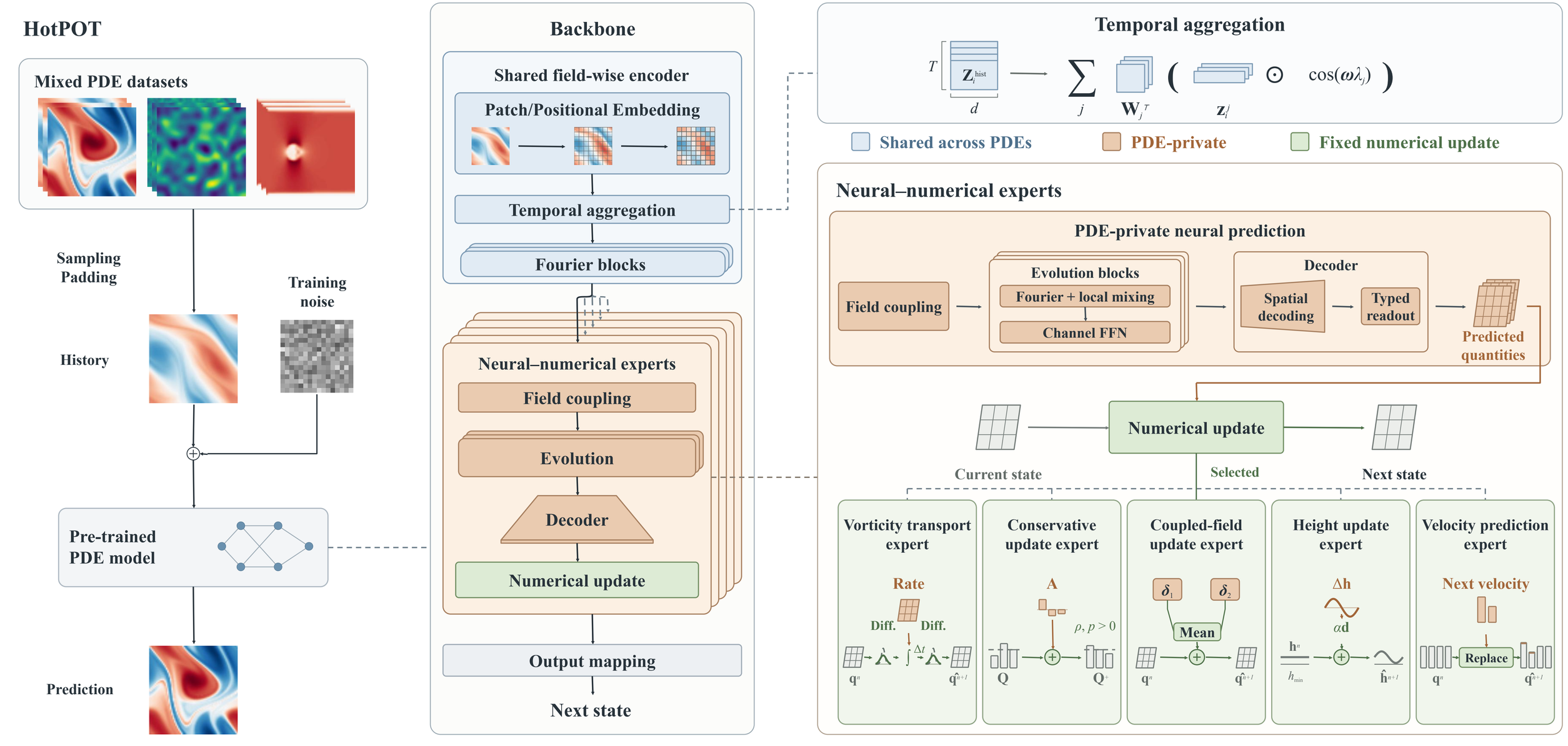

HotPOT
Mixed PDE datasets
Sampling
Padding
History
Training
noise
Pre-trained
PDE model
Prediction
Backbone
Shared field-wise encoder
Patch/Positional Embedding
Temporal aggregation
Fourier blocks
Neural–numerical experts
Field coupling
Evolution
Decoder
Numerical update
Output mapping
Next state
Temporal aggregation
Shared across PDEs
PDE-private
Fixed numerical update
Neural–numerical experts
PDE-private neural prediction
Field coupling
Evolution blocks
Fourier + local mixing
Channel FFN
Decoder
Spatial
decoding
Typed
readout
Predicted
quantities
Numerical update
Current state
Selected
Next state
Vorticity transport
expert
Rate
Diff.
Diff.
Conservative
update expert
A
ρ, p > 0
Coupled-field
update expert
Mean
Height update
expert
Δh
αd
Velocity prediction
expert
Next velocity
Replace